Uncertainty-Aware Machine Learning for Microwave-Based Body Composition Assessment

MASTER'S THESIS

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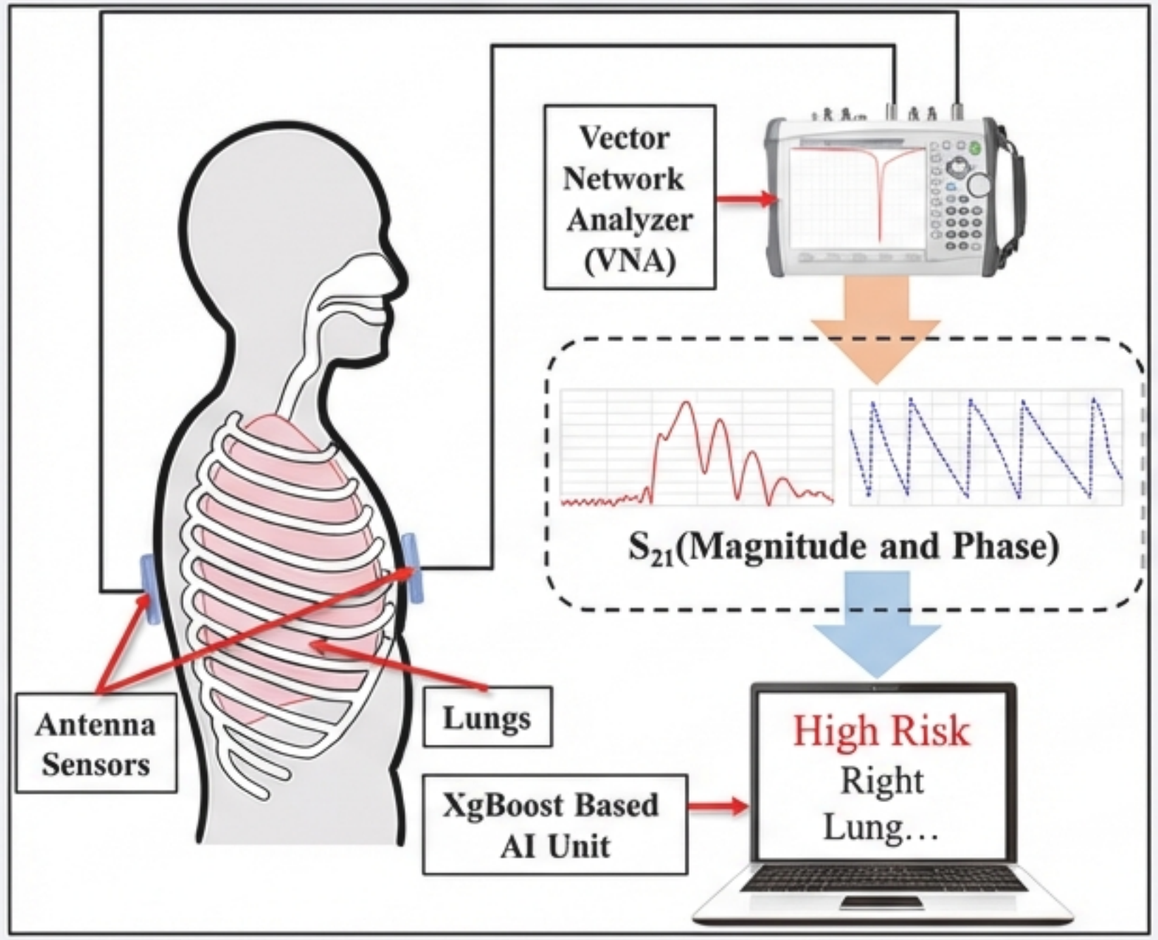
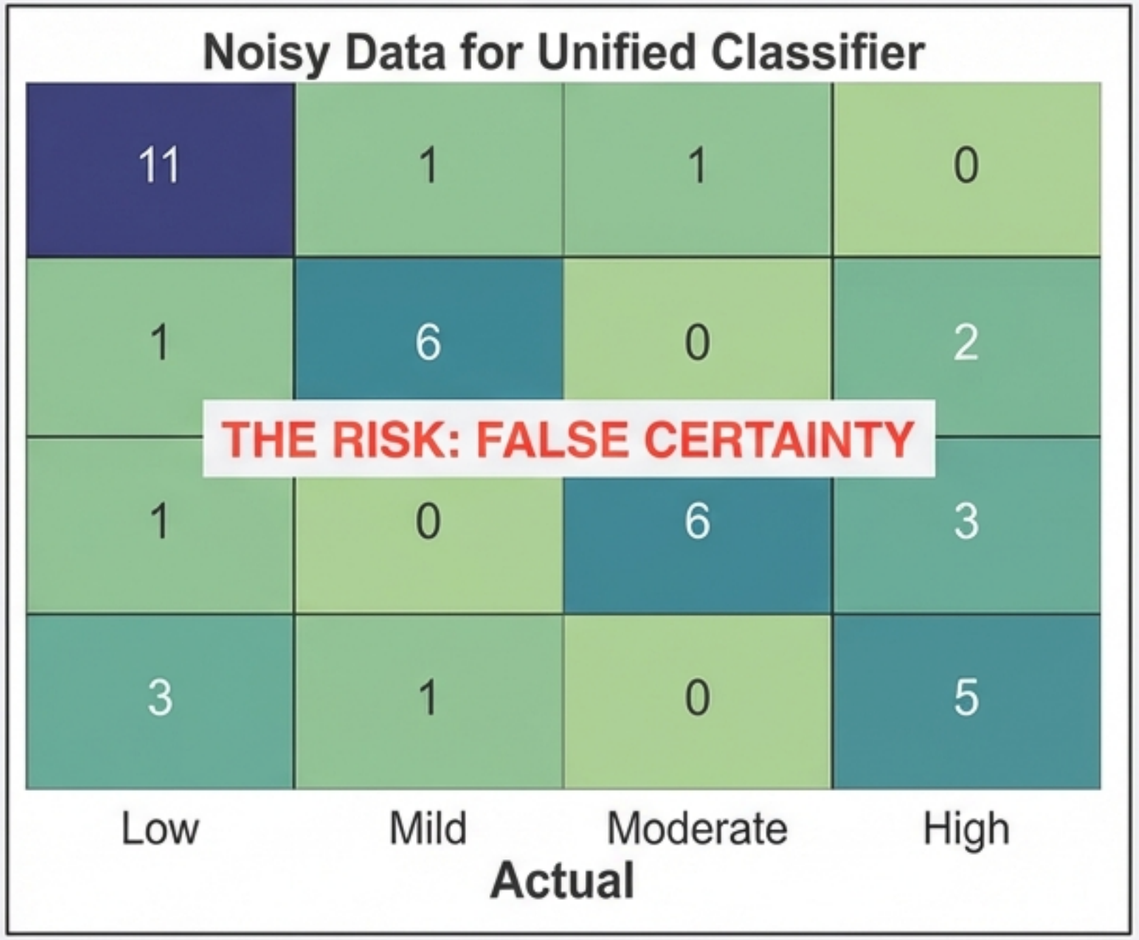
NotebookLM

The ‘Trust Gap’ in Clinical AI

Physical Reality



Data Risk



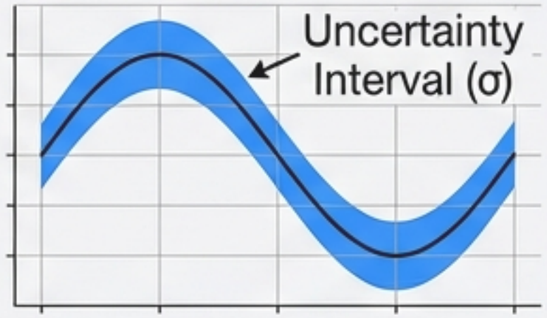
- **Current State:** Microwave Sensors + Deterministic XGBoost
- **The Problem:** Point estimates hide model doubt
- **Clinical Need:** System must flag uncertain predictions

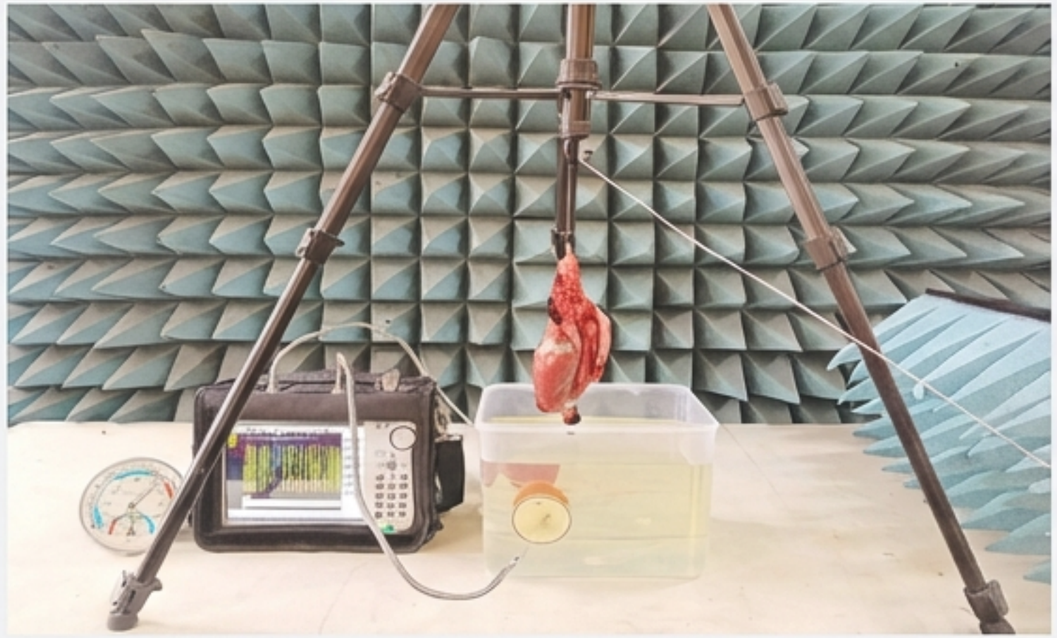
Methodology: From Black Box to Bayesian

Baseline:
XGBoost

Proposed:
Deep Ensembles

Output: Prediction + Confidence





Data Sources:

- Homogeneous Liquid Phantoms
- Ex Vivo Porcine Respiratory Organs
- Volunteer Measurements

Expected Outcome: Enabling Safe Clinical Translation

- Deliverable: Clinical Risk Score
- Mechanism: Automated flags for high uncertainty
- **Impact:** Bridges the gap between lab capability and patient safety